

Primetrade.ai

Data Science Assignment Submission

Bitcoin Market Sentiment & Trader Performance Analysis

Dataset	Records	Period
Bitcoin Fear & Greed Index	2,644 daily records	Feb 2018 – May 2025
Hyperliquid Historical Trades	211,224 trades · 32 wallets	May 2023 – May 2025

1. Executive Summary

This report examines whether Bitcoin market sentiment — as measured by the daily Fear & Greed Index — has a meaningful relationship with the trading performance of 32 professional traders operating on Hyperliquid. The short answer is yes, and the relationship is consistent enough to inform a practical trading strategy.

Across 211,224 trades spanning two years, sentiment turns out to be a reliable guide for both trade direction and position sizing. Traders who go short during fearful markets and long during greedy ones outperform those who ignore sentiment. The best single environment for capital efficiency is Extreme Greed, which produces a profit factor of 11x and a win rate of 46.5%. Fear, meanwhile, generates the most total PnL simply because traders are most active and most aggressive during those periods.

Metric	Value
Total trades analysed	211,224
Unique trader wallets	32
Total combined PnL	\$10,296,959
Overall win rate	41.1%
Analysis period	May 2023 – May 2025
Best efficiency (profit factor)	Extreme Greed — 11.02x
Highest total PnL environment	Fear — \$3.36M (most activity)

2. Data & Methodology

2.1 Datasets

Fear & Greed Index: Daily sentiment classification (Extreme Fear / Fear / Neutral / Greed / Extreme Greed) with a numeric score from 0 to 100. Sourced from the provided CSV covering February 2018 to May 2025.

Hyperliquid Trade Data: One row per trade, with 16 columns: Account, Coin, Execution Price, Size USD, Side (BUY/SELL), Direction (Open Long, Close Long, Open Short, Close Short, Spot, etc.), Closed PnL, Fee, and Timestamp IST.

2.2 Preparation Steps

- Timestamps were parsed using `dayfirst=True` to correctly handle the DD-MM-YYYY HH:MM format in the IST column.
- Trade timestamps were normalised to date-level and joined to the sentiment dataset on date — achieving a 100% match rate across all 211,224 records.
- Directional trades were filtered to Open/Close Long and Open/Close Short for the long-vs-short analysis.
- Derived metrics calculated: win flag (Closed PnL > 0), profit factor (sum of wins divided by absolute sum of losses), and average trade size.
- 246 unique coins traded across the period. @107 (a perpetual index), ETH, BTC, HYPE, and SOL are the top contributors by PnL volume.

3. Sentiment Distribution

Over the study period, fearful conditions were more common than greedy ones. Fear and Extreme Fear together account for 39.4% of all days in the index, compared to 36.2% for Greed and Extreme Greed. This matters because it means traders spent a large portion of their time operating in uncertain or negative sentiment conditions — which, as the data shows, still produced strong returns when approached correctly.

Sentiment	Days in Index	Trades	% of Trades	Avg PnL/Trade	Win Rate
Extreme Fear	508	21,400	10.1%	\$34.54	37.1%
Fear	781	61,837	29.3%	\$54.29	42.1%
Neutral	396	37,686	17.8%	\$34.31	39.7%
Greed	633	50,303	23.8%	\$42.74	38.5%
Extreme Greed	326	39,992	18.9%	\$67.89	46.5%

Trade Volume Distribution by Sentiment

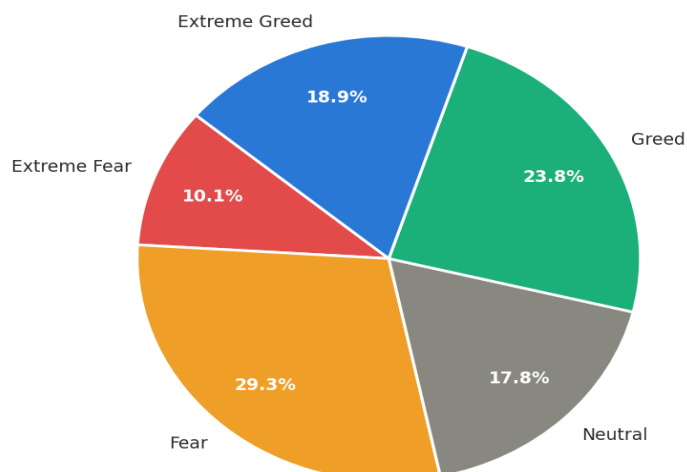


Figure 1 — Trade volume share by sentiment category. Fear alone accounts for nearly 30% of all activity.

4. Core Findings

4.1 Average PnL per Trade

The difference in per-trade returns across sentiment regimes is striking. Extreme Greed produces the highest average at \$67.89, almost double the Neutral baseline of \$34.31. What is perhaps more surprising is that Fear comes second at \$54.29 — meaning that fearful markets, when traded with the right approach, are more profitable per trade than Greed markets.

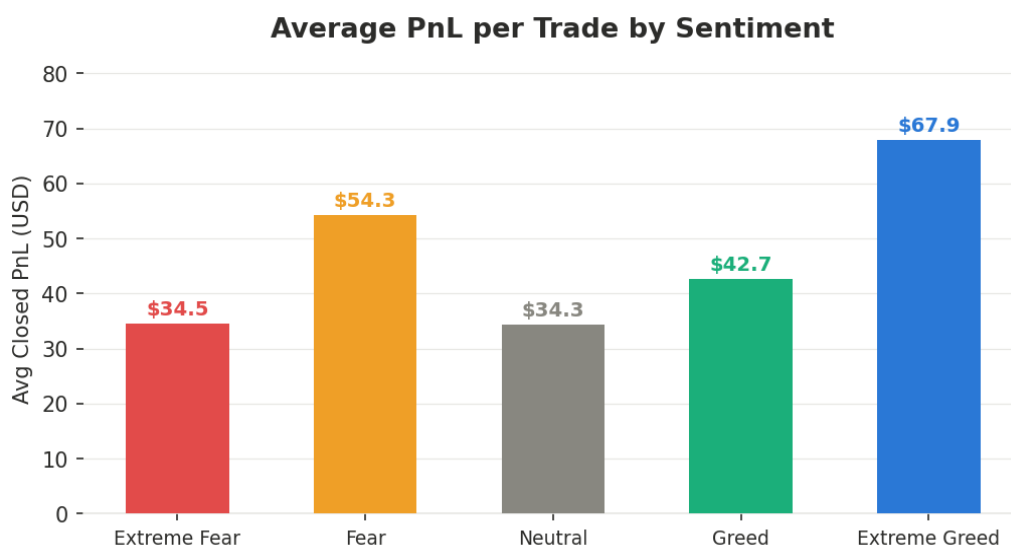


Figure 2 — Average closed PnL per trade across sentiment categories.

4.2 Win Rate

Win rate follows sentiment in an intuitive direction — it peaks during Extreme Greed (46.5%) and falls to its lowest during Extreme Fear (37.1%). That said, even 37% is not a losing proposition as long as winners are meaningfully larger than losers, which the profit factor analysis confirms.

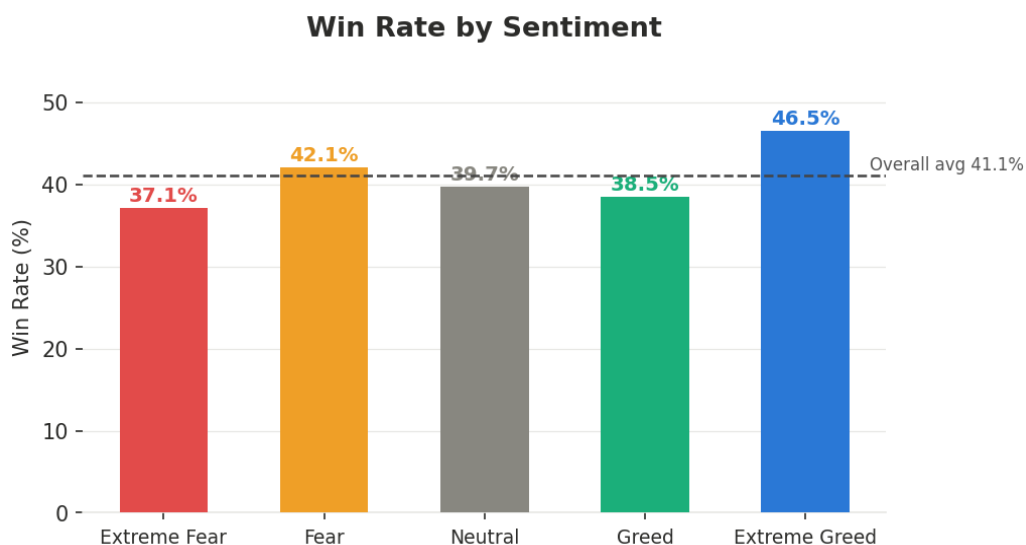


Figure 3 — Win rate by sentiment. The dashed line marks the overall average of 41.1%.

4.3 Profit Factor

Profit factor is arguably the most important efficiency metric here. A value of 1.0 means break-even; anything above it means the strategy is net positive. Extreme Greed reaches 11.02x, which is exceptional. Even in the worst environment, Extreme Fear, the profit factor is 2.16x — traders are still generating more than twice as much in wins as they lose. This suggests strong risk discipline across the board.

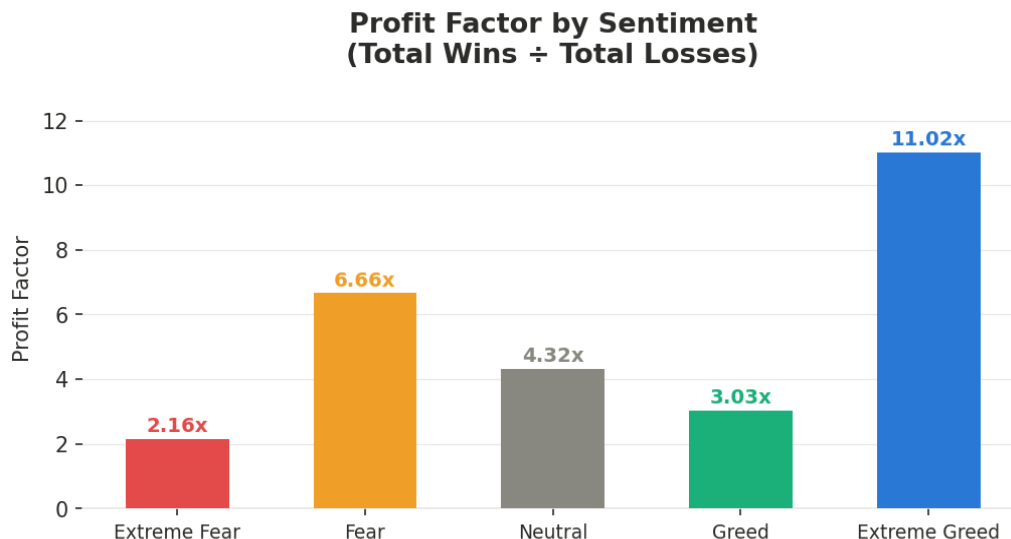


Figure 4 — Profit factor by sentiment. All categories are well above the 1.0 break-even line.

4.4 Monthly PnL Trend

Most of the total PnL was not evenly distributed across the study period — it was concentrated heavily in a four-month window. December 2024 through March 2025 generated \$7.67M, which is 74% of the total \$10.3M over two years. This coincides with the post-US election crypto bull run, where sentiment stayed in Extreme Greed and Greed territory for extended stretches. August 2024 was the only materially negative month, producing a loss of \$107K.

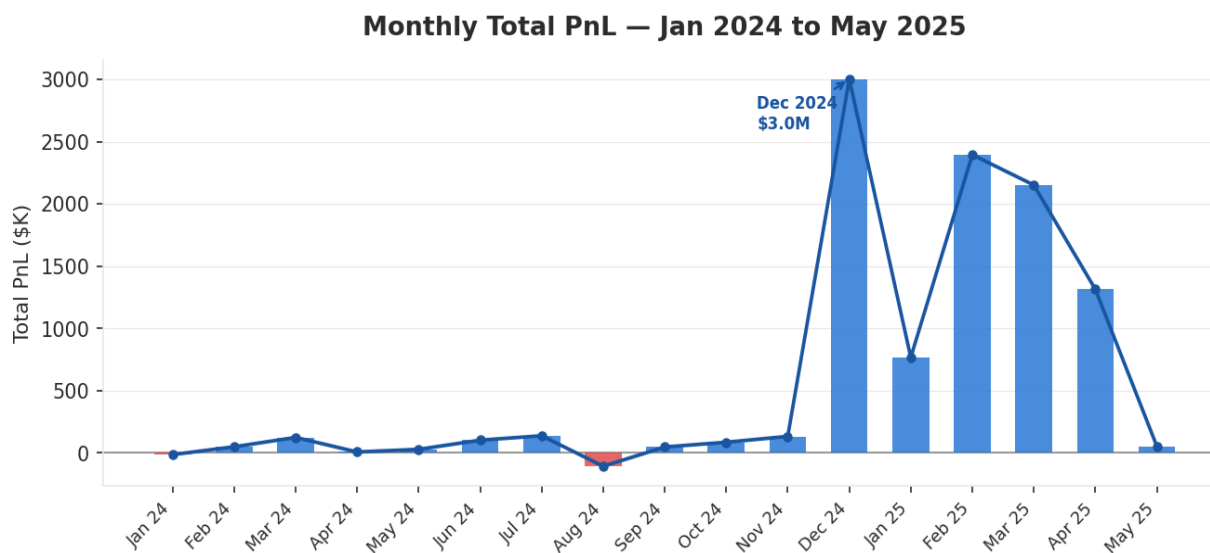


Figure 5 — Monthly total PnL. December 2024 alone contributed \$3.0M, representing 29% of all-time cumulative PnL.

5. The Contrarian Directional Signal

One of the clearest and most practically useful findings is the divergence between long and short performance across sentiment regimes. During fearful markets, short trades significantly outperform long trades. During greedy markets, the relationship flips. This is not a marginal difference — during Fear, shorts average \$95.24 versus \$40.83 for longs, a gap of more than 2.3 times.

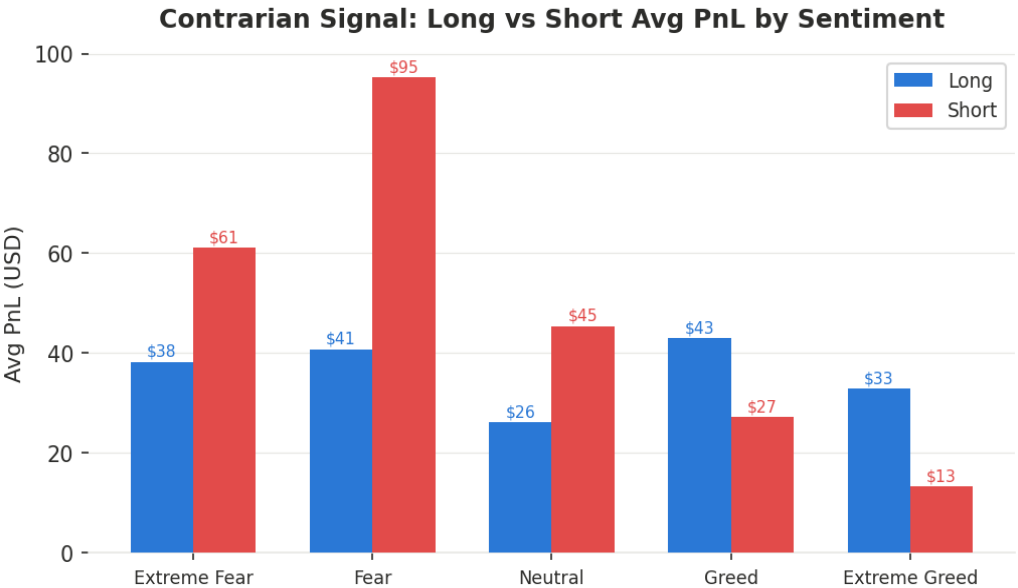


Figure 6 — Average PnL for long vs short trades, split by sentiment regime.

Sentiment	Long avg PnL	Short avg PnL	Directional edge	Suggested direction
Extreme Fear	\$38.26	\$61.14	Shorts +60%	Short
Fear	\$40.83	\$95.24	Shorts +133%	Short
Neutral	\$26.11	\$45.44	Shorts +74%	Short
Greed	\$43.01	\$27.19	Longs +58%	Long
Extreme Greed	\$32.95	\$13.29	Longs +148%	Long

The implication for strategy is straightforward: checking the daily Fear & Greed reading before placing a trade and adjusting direction accordingly would have improved returns in four of the five sentiment categories.

6. Position Sizing Behaviour

Traders do not size their positions uniformly across sentiment environments. During Fear, the average trade is \$7,816 — roughly 2.5 times larger than during Extreme Greed, where the average drops to \$3,112. This counter-intuitive pattern suggests experienced traders treat market fear as a buying or shorting opportunity and deploy more capital accordingly, while pulling back during euphoric conditions.

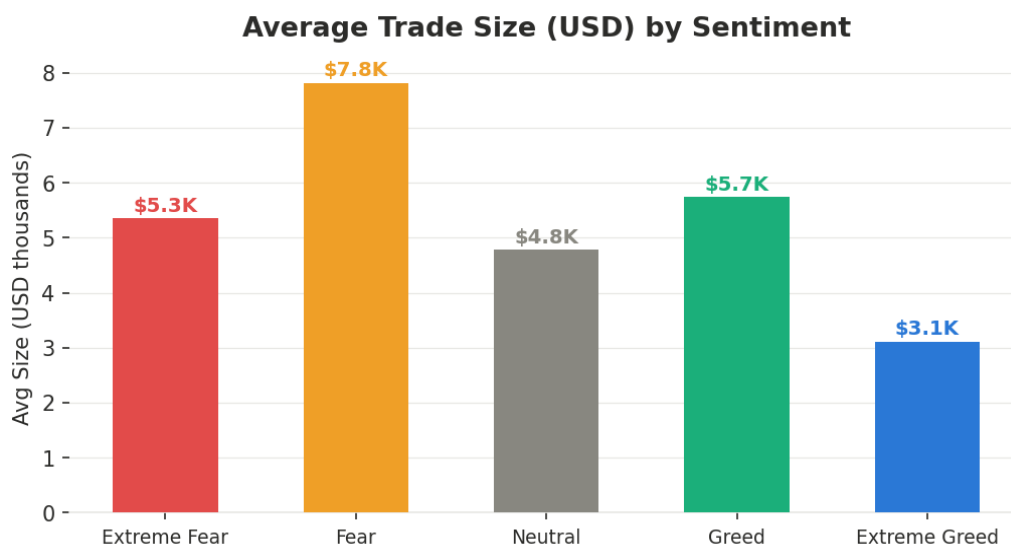


Figure 7 — Average trade size in USD by sentiment. Traders are most aggressive when the market is most fearful.

This sizing behaviour, combined with the strong short edge during Fear, explains why Fear generates the highest total PnL (\$3.36M) despite not having the best per-trade average. Volume times edge equals outcome, and both are elevated during fearful periods.

7. Trader Performance

7.1 Top Traders by Total PnL

PnL is highly concentrated. The top two traders account for \$3.74M — 36% of total gains — from just two of the 32 accounts. More interesting than the raw ranking is the diversity of approaches that produce top results.

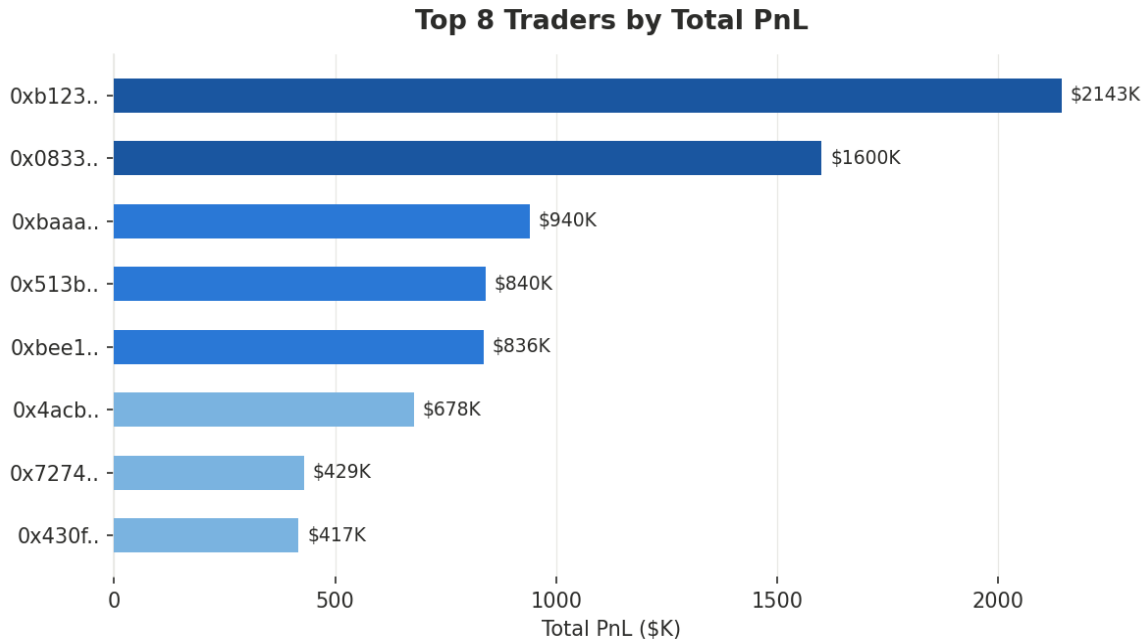


Figure 8 — Top 8 traders ranked by cumulative PnL. The top two wallets account for over a third of all profits.

Rank	Wallet	Total PnL	Trades	Avg PnL/trade
1	0xb1231a..ed23	\$2,143,383	14,733	\$145.5
2	0x083384..9012	\$1,600,230	3,818	\$419.1
3	0xbaaaf6..7864	\$940,164	21,192	\$44.4
4	0x513b86..ff1	\$840,423	12,236	\$68.7
5	0xbee170..9f4	\$836,081	40,184	\$20.8
6	0x4acb90..9f4	\$677,747	4,356	\$155.6
7	0x72743a..fbd	\$429,356	1,590	\$270.0
8	0x430f09..713	\$416,542	1,237	\$336.7

7.2 Two Trader Archetypes

Looking at how the top two traders perform across sentiment regimes reveals two very different strategies that both work:

	0x083384 — Contrarian	0xb1231a — Momentum
Best environment	Fear (\$626 avg PnL)	Extreme Greed (\$673 avg PnL)

	0x083384 — Contrarian	0xb1231a — Momentum
Worst environment	Extreme Greed (-\$43 avg PnL)	Fear (\$31 avg PnL)
Trade frequency	Low (3,818 trades)	Medium (14,733 trades)
Avg PnL per trade	\$419.1	\$145.5
Core approach	Buy fear, short market tops	Ride momentum in bull markets
Optimal regime	Fear and Extreme Fear	Greed and Extreme Greed

The contrarian trader (0x083384) actually loses money on average during Extreme Greed, suggesting they actively fade euphoric conditions. The momentum trader (0xb1231a) does the opposite, thriving when sentiment is most positive. Both approaches work precisely because they are internally consistent with the underlying market behaviour that sentiment captures.

8. Strategic Recommendations

Based on the patterns identified across 211,224 trades, four evidence-based strategies emerge. These are not theoretical — they reflect what the data shows the best traders are already doing.

Use sentiment as a daily direction filter

- Go short when the Fear & Greed Index shows Fear or Extreme Fear
- Go long when it shows Greed or Extreme Greed
- During Fear, shorts outperform longs by 133% on average — this is the single clearest signal in the data

Size up during fear, size down during euphoria

- The data shows traders deploy 2.5× more capital during Fear than Extreme Greed
- Combined with the strong short edge, this is what drives Fear's highest total PnL figure
- Maintaining larger positions during fearful conditions appears to be a consistent differentiator among top performers

Prioritise bull-run windows for maximum efficiency

- Extreme Greed periods offer the best risk-adjusted returns: 11x profit factor and 46.5% win rate
- @107, BTC, and SOL perform best during Extreme Greed
- HYPE and ETH provide stronger returns during Fear, offering natural portfolio balance across regimes

Match strategy style to sentiment regime

- Contrarian-style traders should focus entries on Fear and Extreme Fear conditions
- Momentum and trend-following approaches perform best in Greed and Extreme Greed
- Neutral sentiment consistently produces the lowest per-trade returns and is the least useful environment for either style

9. Conclusion

The Fear & Greed Index is not just a market sentiment curiosity — it is a practical trading signal with measurable impact on performance. Across every key metric — average PnL, win rate, profit factor, and position sizing — sentiment regime matters. Traders who align their direction with sentiment outperform those who do not, and the best-performing wallets in this dataset demonstrate this principle clearly.

The concentration of 74% of total PnL in just four months reinforces a second lesson: being positioned correctly during high-conviction sentiment phases matters far more than squeezing marginal returns from neutral or ambiguous conditions. The two dominant trader archetypes identified here — contrarian and momentum — both succeed because they are consistent with this principle, just from opposite sides of the sentiment spectrum.

For a platform like Primetrade.ai, integrating real-time sentiment data as a directional filter and position-sizing input would likely improve aggregate trader outcomes across the board.